MILITARY SPECIFICATION

ALUMINUM HYDROXIDE GEL WITH MAGNESIUM TRISILICATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Aluminum Hydroxide Gel with Magnesium Trisilicate (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (See 6.3)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a stable suspension of 0.5 gram of Magnesium Trisilicate, USP, added to each 4 ml of Aluminum Hydroxide Gel, USP, with adequate preservative(s) to prevent mold growth.

3.1.1 The material shall be such that when some of the contents of the bottle are removed by pouring and the bottle is recapped, no creeping and growth of filamentous fibers shall occur at the closure.

3.2 Assay. The suspension shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amounts of aluminum hydroxide gel and of magnesium trisilicate when assayed according to the USP monograph for Magaldrate (appropriately modified when necessary).

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3.3 Flavor and palatability. The suspension shall be palatable and pleasant to the taste with no unpleasant after-taste. Not later than the time specified for opening the bids or receipt of proposals, the offeror shall submit to the contracting officer six (6) 12 fl. oz. units of product which are representative of the product which the offeror proposes to furnish. A portion of the samples will be subjected to panel testing for a determination of palatability (see 4.5.1). A portion will be used by the cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determines that such product was previously procured and/or tested by the purchasing activity and found to comply with the specification requirement.

3.4 Ingredients.

3.4.1 Aluminum hydroxide gel. The aluminum hydroxide gel used in the manufacture of the suspension shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.4.2 Magnesium trisilicate. The magnesium trisilicate used in the manufacture of the suspension shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.5 Other ingredients. All other ingredients entering into the finished suspension shall be of USP of NF quality or, if not included in either of these compendia, the ingredients shall be suitable for use in this preparation.

3.6 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the product to the date of delivery to the Government.

3.7 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of suspension produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Immediate container
Minor	II	2.5	Immediate container

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4.4 Examination. The finished suspension shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE II. Classification of defects

Categories	Defects 1/
Major	
101	Suspension not free from foreign matter.
102	Suspension has crystal formation.
103	Suspension not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Tests.

4.5.1 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted), coded, and served to panel members under controlled serving conditions, e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the palatability standard, 1/ when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the samples shall be equal to or greater than the average rating of the standard, similary prepared and tested.

1/ The palatability standard is available, upon separate request to the contracting officer, Defense Personnel Support Center, Directorate of Medical Materiel.

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1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

4.5.2 Tests. Tests shall be performed to determine compliance with all requirements of this document and all referenced documents.

4.6 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One bottle containing 12 fl. oz., as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

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5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged in a corrugated fiberboard box of 200 lb. test "C" flute. Box shall be fitted with a liner, bottom pad and cells. Liner and pad shall be 125 lb. test "C" flute corrugation. The cells shall be 200 lb. test B/C flute, double wall.

5.2.3 Level C. Units shall be packaged in standard commercial container of the size and kind commonly used which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (See 6.1).

5.3.1 Level A. Twenty-four (24) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four (24) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commerical practice.

(b) the National Stock Number.

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NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend: "DO NOT PERMIT TO FREEZE".

5.4.3 Exterior container. Exterior container shall be marked in accordance with 5.5.4 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend: "DO NOT PERMIT TO FREEZE".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock Number	Item Identification
6505-00-221-3415	ALUMINUM HYDROXIDE GEL WITH MAGNESIUM TRISILICATE, 12 fl. oz. (355 ml)

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1634

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-A-37476 ALUMINUM HYDROXIDE GEL WITH MAGNESIUM TRISILICATE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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